Avi Gobrin

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Education

University of Toronto 2023–Present
Bachelor of Applied Mathematics Specialist, Statistics Major, Computer-Science Minor *Toronto, ON*

Relevant Coursework
Real Analysis, Nonlinear Optimization, Probability and Statistics I/II, Multivariate Data Analysis, Statistical Computation, Algorithm Design and Analysis, Graph Theory, Combinatorics

Experience

Nextdoor May 2026 – Present
Machine Learning Software Engineering Intern *San Francisco, CA*

- Designed and shipped an end-to-end system ranking neighborhoods by family-friendliness, turning an ambiguous, open-ended research question into a deployed product feature.
- Owned the full stack, from a data-processing layer ingesting heterogeneous researched signals, through a backend scoring service combining them into a single interpretable index, to the frontend surfacing ranked recommendations, validating ranking quality against a dedicated evaluation harness before launch.

Shopify May–Aug 2025
Software Engineering Intern *Toronto, ON*

- Increased new-merchant activation and retention, routing roughly 100 merchants/day through guided onboarding, by launching a partner-attach system that automatically pairs each new merchant with a vetted onboarding advisor.
- Brought German VAT submissions into compliance with updated tax regulations, unblocking hundreds of affected merchants and reducing compliance risk, by integrating the German tax-code API and aligning Shopify's existing tax flows with the new legal requirements.
- Improved system reliability and reduced recurring issues by resolving cross-disciplinary technical debt assigned by adjacent teams, collaborating with engineering and product stakeholders to ship scalable fixes.

Projects

Multivariate Analysis of Student Outcomes (OULAD), Independent Research *2026*

- Recovered latent engagement and performance structure across 23,301 student-course records via a 3-component PCA basis (72.9% variance) and varimax factor model, after formally rejecting multivariate normality (Mardia, Box's M).
- Showed PCA, factor analysis, GMM, and LDA converge on one two-dimensional structure; benchmarked classifiers under 5-fold CV (logistic regression 67.0% vs. 53.0% baseline).

Autonomous Planning & School Knowledge Agents, Personal Project *2026*

- Built an autonomous daily-planning agent (Claude Code, Obsidian) that runs on its own at the start and end of each day: every evening it reviews what I completed, writes structured notes on the day, then regenerates and time-blocks the next day's calendar from a parsed task file and a traffic-light priority scheme.
- Built a companion school-workflow agent that auto-formats lecture and problem-set notes and links related notes into a connected knowledge base of theorem references and cross-links, surfacing relevant material to accelerate homework, projects, and exam preparation.

F1 Race Winner Predictor, Personal Project *2024*

- Built a supervised pipeline predicting Formula 1 race winners from driver/constructor rolling form, qualifying results, track characteristics, and weather.
- Engineered features such as driver–track history and pit-stop reliability from historical race and weather data. Evaluated on held-out races, the classifier reached 40% race-winner and 60% podium-finish accuracy.

Technical Skills

Languages Python, R, SQL, Java, JavaScript
Quantitative/Data Probability & Statistics, Optimization, Multivariate Analysis; NumPy, Pandas, scikit-learn, Databricks, Git
AI Tools Claude, Claude Code, Cursor, Hermes, LLM Agents